

Cheng Chen

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I work on world models and robot learning from heterogeneous experience, with hands-on large-video-model training and data work. I am interested in how general knowledge, robot demonstrations, and deployment experience contribute differently to generalization.

EDUCATION

Nanyang Technological University, Singapore | Jan. 2023 - present

Ph.D. in Computer Science. Supervisors: Prof. Guosheng Lin and Dr. Fayao Liu.

Defense expected December 2026.

Fudan University, Shanghai, China

M.Eng. in Information Engineering, 2019 - 2022; B.Eng. in Electronic Engineering, 2015 - 2019.

SELECTED RESEARCH CONTRIBUTIONS

AnyWorld | First author | CoRL 2026

Factorized egocentric world modeling for experience recomposition across embodiment, camera, and scene context, with target-specific action calibration. Team evaluation: IRON grasping improved from 4/20 to 11/20 trials with generated data.

[Project and real-robot examples](#).

XPACE | Co-author | 2026 preprint

Joint world and action modeling from heterogeneous human and robot experience. My contribution: world-model training and data. [Project and real-robot examples](#).

RoboGPT | Ongoing research

Investigating why general-purpose agents generalize in robotics. Testing whether task-irrelevant interactions reveal embodiment knowledge, whether task and embodiment experience can be composed, and where adaptation fails under changed action semantics or limited physical feedback.

SwiftVLA | Co-first author | CVPR 2026

Lightweight vision-language-action modeling with spatiotemporal representations for efficient robot action generation.

RESEARCH EXPERIENCE

XPENG Robotics | Research Intern | Dec. 2025 - present

World models and robot learning, collaborating with Jiacheng Wei, Jiawang Bai, and Xiao-Yang Guo.

Nanyang Technological University | Ph.D. researcher | Jan. 2023 - present

3D generation, video generation, world models, and robot learning.

Ant Group | Research Intern | Apr. 2025 - Oct. 2025

Video generation, video editing, and interactive generative video models.

Earliest start: February 2027, subject to degree completion and work authorization.

SELECTED PUBLICATIONS AND PREPRINTS

AnyWorld: Factorized Egocentric World Models for Cross-Embodiment Generalization | First author

Cheng Chen et al. *CoRL* 2026.

XPACE: Joint World and Action Modeling from Heterogeneous Experience | Co-author

Jiacheng Wei, Jerry Bai, Xiaoyu Yue, Zidong Wang, Xiaoyang Guo, Cheng Chen, et al. *arXiv preprint*, 2026.

SwiftVLA: Unlocking Spatiotemporal Dynamics for Lightweight VLA Models at Minimal Overhead | Co-first author

Chaojun Ni, Cheng Chen (co-first author), et al. *CVPR* 2026.

DiVA: Enabling Interactive Digital Life Simulation via Video Models | First author

Cheng Chen et al. *arXiv preprint*, 2026; *under review at IJCV*.

CADCrafter: Generating Computer-Aided Design Models from Unconstrained Images | First author

Cheng Chen et al. *CVPR* 2025.

Text-to-Image Rectified Flow as Plug-and-Play Priors | Co-author

Xiaofeng Yang, Cheng Chen, Xulei Yang, Fayao Liu, Guosheng Lin. *ICLR* 2025.

Sculpt3D: Multi-View Consistent Text-to-3D Generation with Sparse 3D Prior | First author

Cheng Chen et al. *CVPR* 2024.

Learn to Optimize Denoising Scores: A Unified and Improved Diffusion Prior for 3D Generation | Co-author

Xiaofeng Yang, Yiwen Chen, Cheng Chen, et al. *ECCV* 2024.

UTC: A Unified Transformer with Inter-Task Contrastive Learning for Visual Dialog | First author

Cheng Chen et al. *CVPR* 2022.

AWARDS AND HONOURS

SINGA Scholarship, Singapore International Graduate Award	Jan. 2023
National Scholarship (1/72)	Oct. 2021
Outstanding Graduate of Fudan University	Apr. 2022
First-Class Scholarship for Outstanding Students, Fudan University	Oct. 2019
Outstanding Student, Information School of Fudan University	May 2017

SKILLS

Training: Full-parameter fine-tuning of 10B+ video models; multi-GPU distributed training; mixed precision; data and evaluation pipelines.

Generative modeling: Diffusion and flow matching; diffusion transformers; image/video generation and editing; 3D generation.

Robotics: Video-action world models; VLA; policy learning and adaptation; action calibration; real-robot closed-loop evaluation.

Tools: Python, PyTorch, TensorFlow, Linux, LaTeX. **Languages:** Mandarin Chinese, English.